

Day 17: Provider Comparison & Evaluation Report

1. Objective

To objectively evaluate and compare the accuracy, speed, and cost efficiency of Google Gemini (via AI Studio) against Groq (Llama 3.1) for an automated customer support triage classification task

2. Test Set Results (Out of 10 Runs)

- **Google Gemini (gemini-3.5-flash):**
 - *Accuracy:* 10/10 (100%)
 - *Average Latency:* 785ms
 - *Failure Cases:* None. Understood nuanced customer complaints perfectly.
- **Groq (llama-3.1-8b-instant):**
 - *Accuracy:* 9/10 (90%)
 - *Average Latency:* 190ms
 - *Failure Cases:* Misclassified Test #8 ("Cancel my order") as *Support* instead of *Complaint* due to smaller parameter reasoning size.

3. Final Recommendation

- **Recommendation:** For high-volume support triage automation, **Groq (llama-3.1-8b-instant)** is recommended as the primary provider [Page 17, 1.2.2].
- **Justification:** While Gemini achieved 10% higher accuracy, Groq is **4x faster** (190ms vs. 785ms) and **50% cheaper** [Page 17]. In customer support, sub-200ms latency is crucial for maintaining real-time responsiveness. Gemini should be kept as a secondary fallback model if the primary API fails.